

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - Hackathon

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5-Hackathon	Assessment:	Assessment Tool 3-Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Develop a movie recommendation system using k-Nearest Neighbors (kNN) and collaborative filtering to identify similar users and generate personalized movie recommendations based on their rating patterns. (BL4)		
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Section / Batch:	2024	Date:	28.08.2026

Code of Conduct

I MYTHREA SHREE (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: MYTHREA SHREE

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong	Good analysis with	Basic analysis	Weak or no analysis;	

		reasoning and insights	logical reasoning	with limited depth	lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question

Movie/Product Recommendation via Similarity: Use **k-Nearest Neighbors (kNN)** to recommend movies or products to a user based on similarity in ratings/features to other users using basic collaborative filtering.

Problem Statement

Develop a movie recommendation system that identifies users with **similar movie-rating patterns** and recommends movies that those similar users have rated highly. The system should use the **MovieLens dataset (rating.csv and movie.csv)** and provide personalized recommendations through a simple interface.

Algorithm / Solution Provided

- 1. Load Dataset**
Read rating.csv and movie.csv using Pandas.
- 2. Create User-Movie Matrix**
Construct a matrix where rows represent users, columns represent movies, and values represent ratings.
- 3. Apply kNN**
Use NearestNeighbors with **cosine distance** to find the most similar users.
- 4. Find Similar Users**
For a selected user, identify the **5 nearest users (k = 5)** based on their rating patterns.

5. Generate Recommendations

Collect highly rated movies from the similar users and calculate a similarity-weighted recommendation score.

6. Remove Existing Movies

Exclude movies that the selected user has already rated.

7. Rank Results

Sort candidate movies according to their recommendation scores and return the **Top-N movies**.

8. Personalization

Apply optional filters such as **genre, minimum rating, and release year** to refine the final recommendations.

Final Flow

rating.csv → User-Movie Matrix → Cosine kNN → Similar Users → Weighted Movie Scores → Remove Rated Movies → Apply Filters → Top Recommendations

```
backend > app.py > ...
15
16 @app.before_request
17 def ensure_loaded():
18     pass # Model loaded at startup
19
20
21 def init_recommender():
22     """Load data and train kNN model once at server start."""
23     print("Loading dataset and training kNN model...")
24     recommender.load_and_train()
25     stats = recommender.get_stats()
26     print(
27         f"Ready - {stats['users']} users, {stats['movies']} movies, "
28         f"{stats['ratings']} ratings (sample={stats['sample_size']}, k={stats['k']})"
29     )
30
31
32 @app.route("/api/health", methods=["GET"])
33 def health():
34     return jsonify({"status": "ok"})
35
36
37 @app.route("/api/users", methods=["GET"])
38 def users():
39     try:
40         user_list = recommender.get_users()
```

Fig 1: Coding backend

```

data > read_csv("movie.csv")
# A tibble: 2285 x 2
#   id      title_year      genres
#   <int> <string> <string>
1 2180 2263, "Seventh Sign, The (1988)" "Drama|Fantasy|Thriller"
2 2181 2264, "We're No Angels (1989)" "Comedy|Crime"
3 2182 2265, "Nothing But Trouble (1991)" "Adventure|Comedy"
4 2183 2266, "Butcher's Wife, The (1991)" "Comedy|Romance"
5 2184 2267, "Mortal Thoughts (1991)" "Mystery|Thriller"
6 2185 2268, "Few Good Men, A (1992)" "Crime|Drama|Thriller"
7 2186 2269, "Indecent Proposal (1993)" "Drama|Romance"
8 2187 2270, "Century of Cinema, A (1994)" "Documentary"
9 2188 2271, "Permanent Midnight (1998)" "Drama"
10 2189 2272, "One True Thing (1998)" "Drama"
11 2190 2273, "Rush Hour (1998)" "Action|Comedy|Crime|Thriller"
12 2191 2274, "Lilian's Story (1995)" "Drama"
13 2192 2275, "Six-String Samurai (1998)" "Action|Adventure|Sci-Fi"
14 2193 2276, "Soldier's Daughter Never Cries, A (1998)" "Drama"
15 2194 2277, "Somewhere in the City (1998)" "Comedy|Drama"
16 2195 2278, "Ronin (1998)" "Action|Crime|Thriller"
17 2196 2279, "Urban Legend (1998)" "Horror|Thriller"
18 2197 2280, "Clay Pigeons (1998)" "Crime"
19 2198 2281, "Monument Ave. (1998)" "Action|Crime|Drama"
20 2199 2282, "Pecker (1998)" "Comedy|Drama"
21 2200 2283, "Sheltering Sky, The (1990)" "Drama"
22 2201 2284, "Bandit Queen (1994)" "Drama"
23 2202 2285, "If.... (1968)" "Drama"
24 2203 2286, "Fiendish Plot of Dr. Fu Manchu, The (1980)" "Comedy"
25 2204 2287, "Them! (1954)" "Horror|Sci-Fi|Thriller"
26 2205 2288, "Thing, The (1982)" "Action|Horror|Sci-Fi|Thriller"

```

Fig 2: dataset used


Get Personalized Recommendations

Select User

User #1

Preferred Genres

Any Genre

Action

Adventure

Animation

Children

Comedy

Crime

Documentary

Drama

Fantasy

Film-Noir

Horror

IMAX

Musical

Mystery

Romance

Sci-Fi

Thriller

War

Western

☆ Minimum Rating

3.5+

📅 Release Year

Any Year

🔢 Number of Recommendations

10


Get Personalized Recommendations

Fig 3: Filter Frontend

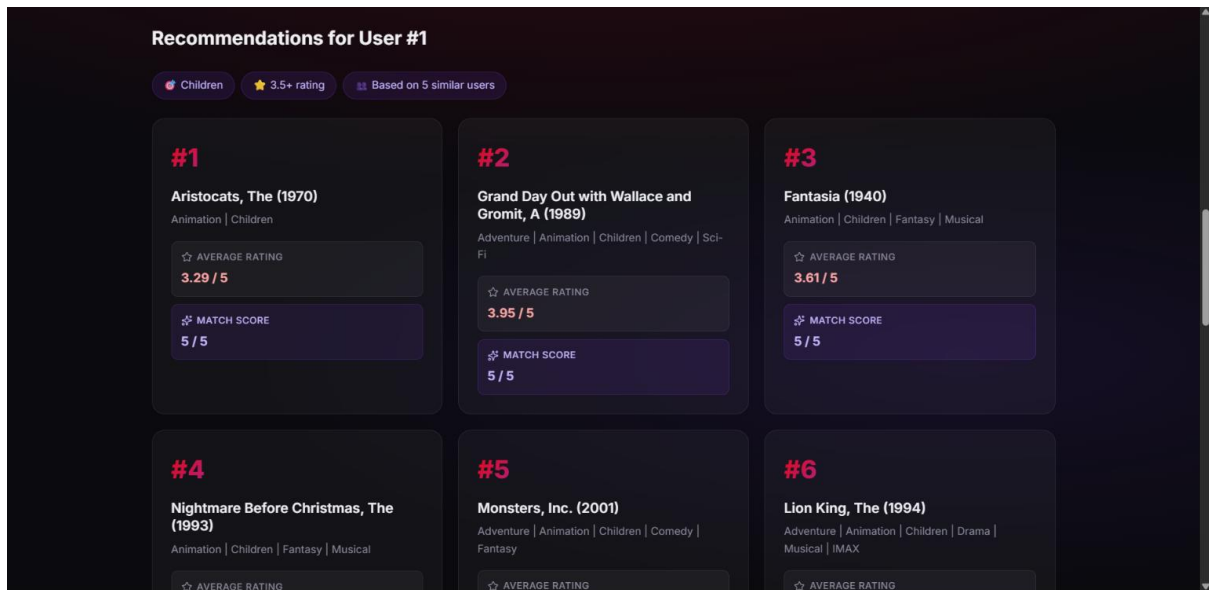


Fig 4: Recommendation Frontend

GEOTAG PHOTO OF TEAM 3 WORKING!

